

DATA 266 Assignment 4

Mini GPT from Scratch

DATA 266 Assignment 4, findings document

1. Personal parameters

Standing requirement 0.1, derived from the last four digits of my SJSU student ID.

SID4	SEED	SLICE	HP_ID	CLS_A	CLS_B
5330	5330	330	2	0	3

`SEED` seeds Python, NumPy and PyTorch in the first cell of the notebook. The other four are stated as required but not referenced by this assignment: the model trains on the full corpus, so there is no subset to index; it is a language model rather than a classifier, so there is no class pair; and Assignment 4 supplies no `HP_ID` mapping, so the arm selects nothing.

2. Overview

A decoder only Transformer built from `nn.Linear`, `nn.LayerNorm`, `nn.GELU` and `nn.Embedding`. The multi head attention, the causal mask, the decoder block and the sampling loop are written by hand. `nn.Transformer`, `nn.MultiheadAttention` and HuggingFace models are not used. All numbers come from one run of `notebooks/assignment4_mini_gpt.ipynb`.

3. Data preparation

The corpus is 1,115,394 characters. Character level tokenization gives a vocabulary of the 65 distinct characters in the file, so encoding is exact and invertible.

Sequence length is 128. A window at position t gives input `[t ... t+127]` and target `[t+1 ... t+128]`, so every one of the 128 positions is a next character prediction. A stride of 32 gives 34,853 windows.

```
input : 'First Citizen:\nBefore we proceed any further, hear me speak.'
target: 'irst Citizen:\nBefore we proceed any further, hear me speak.\n'
```

4. Architecture

Attention. Queries, keys and values come from separate linear projections. The model dimension splits into 4 heads that run in one batched matmul. Scores are scaled by the square root of the head dimension, which keeps the softmax out of its saturated region. The causal mask is applied to the scores with `masked_fill(..., -inf)` before the softmax, so position i attends only up to i ; masking after the softmax would leave rows not summing to one.

Block. Attention, residual, LayerNorm, then Linear $\rightarrow$ GELU $\rightarrow$ Linear at four times the model width, residual, LayerNorm.

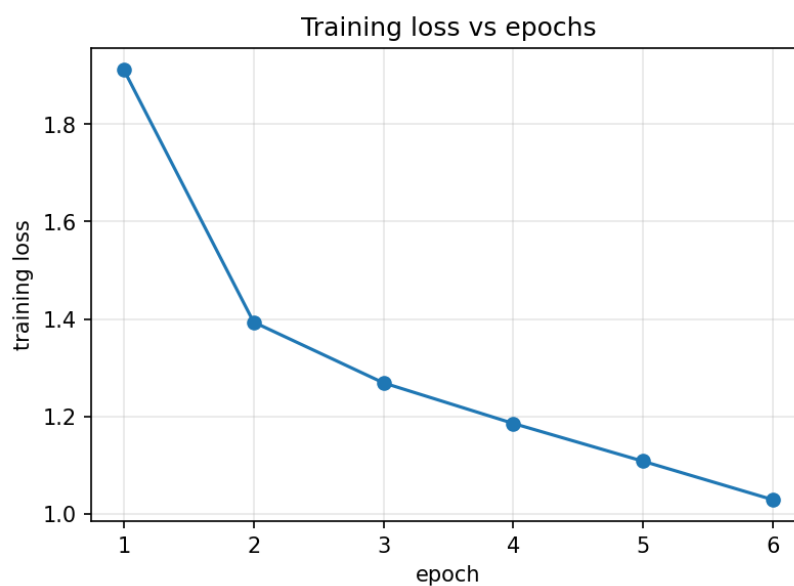
Model. Token embedding plus learnable positional embedding, four blocks, final linear projection to vocabulary logits. The positional embedding is needed because attention alone is permutation equivariant.

Property	Value
Total parameters	3,225,153
Hidden dimension	256
Heads	4
Layers	4
Context length	128
Vocabulary size	65

5. Training

Adam, learning rate 1e-3, batch size 64, 6 epochs.

Epoch	1	2	3	4	5	6
Training loss	1.9114	1.3937	1.2691	1.1858	1.1083	1.0293



Training loss vs epochs

The loss falls from 1.9114 to 1.0293 and is still decreasing at epoch 6, against a uniform baseline of $\ln(65) = 4.17$.

6. Decoding comparison

Each setting is scored on two measures, averaged over three generations of 300 characters from the prompt `ROME0: .` Valid word rate is the fraction of words that occur in the corpus, so it measures

coherence. Unique trigram rate is the fraction of character trigrams that are unique, so it measures diversity.

Decoding	Valid words (coherence)	Unique trigrams (diversity)
greedy	0.952	0.574
temperature 0.5	0.962	0.734
temperature 1.0	0.924	0.831
temperature 1.5	0.770	0.884
top k = 2	0.943	0.730
top k = 40	0.924	0.831

Greedy

Takes the argmax at every step, so it is deterministic. It spells well (0.952) but has by far the lowest diversity (0.574) and visibly degenerates: the sample says “the sea” four times in six lines. With no randomness the model cannot leave a high probability loop once it enters one.

```
ROMEO:
I have not this forgiven with a fortune that
they have seen their file traged witchcry.

MERCUTIO:
Nay, then they say in the senself against the sea,
And then the sea and stand betwixt on the sea,
And then the sea and stand but a sea,
And then to the sea a stain and stir, as I
```

Temperature

Divides the logits by T before the softmax. T below 1 sharpens toward the mode, T above 1 flattens toward uniform. Both measures move monotonically: as T rises from 0.5 to 1.5 the valid word rate falls (0.962, 0.924, 0.770) and diversity rises (0.734, 0.831, 0.884). At T = 0.5 the text stays near the mode and spells almost perfectly. At T = 1.5 unlikely characters are picked often, giving invented words such as *inverext*, *GreWts* and *mixtUr*.

```
ROMEO:
O inverext deoth!
Tybalt should through men,--and

QUEEN MARGARET:

KING HENRY VI:
Must I, brother? wast the more affore, slaves it.

QUEEN ELIZABETH:
After thou dismay'st, the Castle wrecks.
```

Top k

Keeps the k most likely characters and renormalises, deleting the tail rather than reshaping the whole distribution. k = 2 is more coherent than k = 40 (0.943 against 0.924) and much less diverse (0.730 against 0.831). At k = 2 the model picks between two characters per step and drifts into repetition, again ending on the sea.

```
ROMEO:
Not a seal'd wing, and so soundly as this.

BENVOLIO:
I thank you shall charge him, sir, there's
man as yourself as a wife, that you
have man betwixt the seeme he sent a singer
```

k = 40 produced output identical to T = 1.0. With 65 characters in the vocabulary the 25 lowest ranked ones are never sampled anyway, so truncating them changes nothing. Top k only does real work when k is small relative to the number of characters the model is actually considering, which on a vocabulary of 50,000 subword tokens would almost always hold.

7. Conclusion

Most coherent: temperature 0.5, at 0.962. Greedy is close at 0.952 and spells reliably, but it repeats itself so heavily that it reads worse as text. Sampling at low temperature keeps the spelling and avoids the loop.

Most diverse: temperature 1.5, at 0.884, the highest of the six, bought with the worst coherence at 0.770.

Decoding trades coherence against diversity, and the two knobs do it differently. Temperature rescales the whole distribution. Top k truncates its tail and leaves the head untouched.